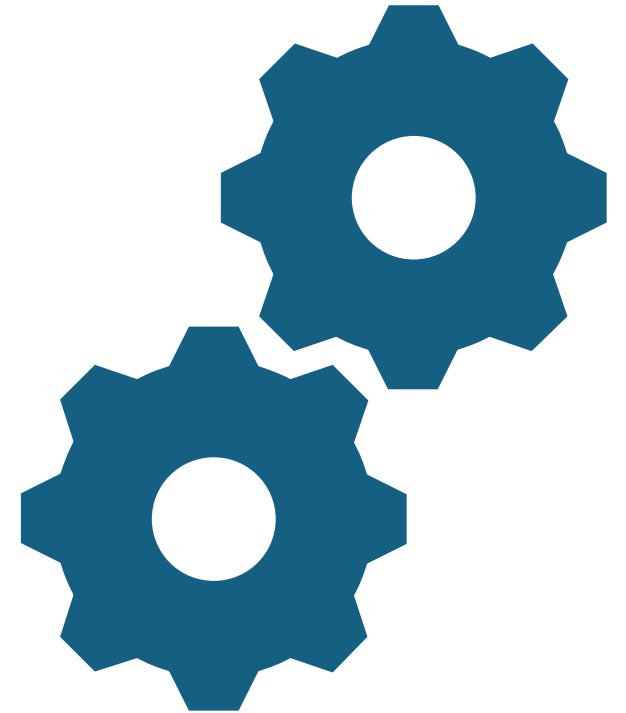


Advanced Deterministic Systems Optimization

ISE 555 Unit 3



Overview

- Part 1: Linear Programming Fundamentals
- Part 2: Converting to Standard Form
- Part 3: Feasible Solutions and Basic Feasible Solutions

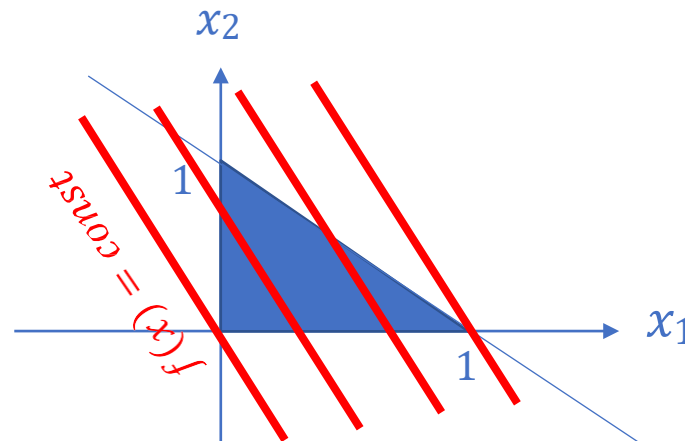
Part 1: Linear Programming Fundamentals



Why do a Geometric Approach?

- In 2-D problems (problems with 2-dimensional x) we can graphically examine the notions of feasibility and optimality
- This allows us to consider potential optimization strategies intuitively
- Many of these same strategies will be employed in the Simplex Method
- Example from first class:

$$\begin{array}{ll}\text{maximize} & f(x) = 2x_1 + x_2 \\ \text{subject to} & x_1 + x_2 \leq 1 \\ & x_1 \geq 0, x_2 \geq 0\end{array}$$



$$(x_1, x_2)^* = (1, 0)$$

Canonical Form for Linear Programs

- We have been using a **canonical form** for linear programs as follows:

$$\begin{array}{ll}\text{minimize} & z = c^T x \\ \text{subject to} & Ax \leq b \\ & x \geq 0\end{array}$$

- Why is this form useful?

A: This is a natural form for formulation practical problems. Many constraints are naturally written as inequalities.

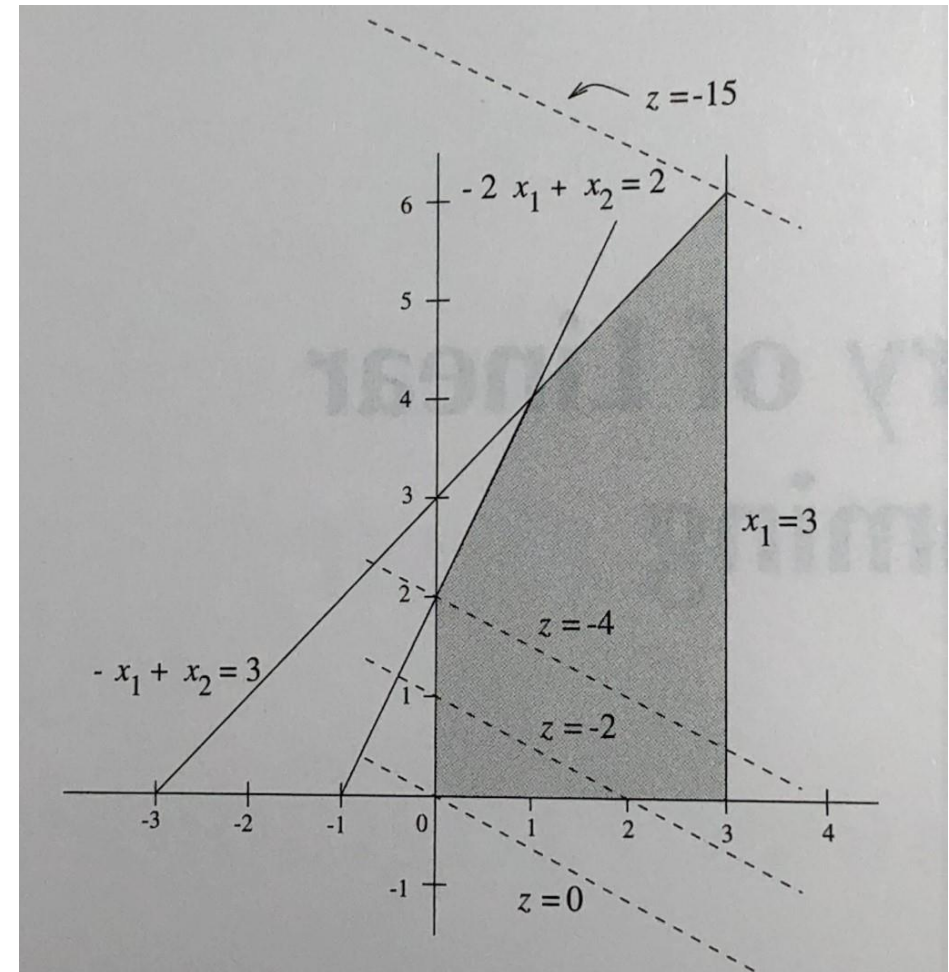
- What are potential downsides of this form?
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Another Example

- Consider the following problem:

$$\begin{array}{ll}\text{minimize} & z = -x_1 - 2x_2 \\ \text{subject to} & -2x_1 + x_2 \leq 2 \\ & -x_1 + x_2 \leq 3 \\ & x_1 \leq 3 \\ & x_1 \geq 0, x_2 \geq 0\end{array}$$

**DISPLAY FEASIBLE REGION
ON WHITE BOARD**



What happened

- The minimum occurred at a corner of the feasible region, why?
 - Where else can it occur?
 - What does constraints being linear imply?
 - What does the objective being linear imply?
 - So, for a linear program (linear objective and constraints) we have that linear programs with a single optimal solution must have that solution exist at an extreme point (corner of feasible region). This is the basis of the Simplex Method that is used by commercial solvers for linear programs.
 - What is a corner?
 - ANSWER: A boundary point that exists on two boundaries (for this 2-D case)
 - More formally corners are known as extreme points. An **extreme point** x of a convex set $\mathcal{S}$ is a point $x \in \mathcal{S}$ that cannot be expressed as a convex combination of two feasible points that are different from x .
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Part 2: Converting to Standard Form



Standard Form for Linear Programs

- We will now consider using a **standard form** for linear programs as follows:

$$\begin{array}{ll}\text{minimize} & z = c^T x \\ \text{subject to} & Ax = b \\ & x \geq 0\end{array}$$

where we require $b \geq 0$

- What is different from our earlier representations?
 - Why is this form useful?
 - A: The equality form of constraints allows us to consider feasibility much more simply, as the only inequality constraints are the lower bound on the variables themselves
 - What are potential downsides of this form?
 - How do we convert from other representations?
-

Converting to Standard Form

- What to do about maximize?

A: Multiple objective by -1

- What if the objective has a constant (ie: $z = 2x_1 + x_2 - 4$)?

A: We can ignore it and optimize $z' = 2x_1 + x_2$

- What if a constraint has $b < 0$?

A: Multiple the constraint by -1 (note that flips any inequalities)

$$\begin{array}{l} \text{minimize} \\ \text{subject to} \end{array} \quad \begin{array}{l} z = c^T x \\ Ax = b \\ x \geq 0 \end{array}$$

- What to do about variables that have non-zero bounds (ie: $x_1 \geq 2$)?

A: Form $x_1' = x_1 - 2$ then $x_1' \geq 0$

- What to do about upper and lower bounds (ie: $x_1 \geq 0$ and $x_1 \leq 7$)?

A: Treat the upper bound as a general constraint

- What to do about unbounded variables (ie: x_2 is unbounded)?

A: Replace with two new variables x_2' and x_2'' with $x_2 = x_2' - x_2''$ and $x_2', x_2'' \geq 0$

- What about inequality constraints of less than (i.e. $x_1 + x_2 \leq 2$)?

A: Add a slack variable s_1 as $x_1 + x_2 + s_1 = 2$ and $s_1 \geq 0$

- What about inequality constraints of greater than (i.e. $3x_1 + 2x_2 \geq -10$)?

A: Add an excess/surplus variable e_1 as $3x_1 + 2x_2 - e_1 = -10$ and $e_1 \geq 0$

A LOT OF THIS WEEK'S ASSIGNMENT WILL BE DOING THESE CONVERSIONS

Example of Conversions

$$\begin{array}{ll}\text{maximize} & z = 2x_1 + x_2 \\ \text{subject to} & x_1 + x_2 \leq 1 \\ & x_1 \geq 0, x_2 \geq 0\end{array}$$



- 1) Flip sign to change max to min: minimize $z' = -2x_1 - x_2$
- 2) Add slack variable to change $\leq$ to $=$: $x_1 + x_2 + s_1 = 1, s_1 \geq 0$



$$\begin{array}{ll}\text{minimize} & z' = -2x_1 - x_2 \\ \text{subject to} & x_1 + x_2 + s_1 = 1 \\ & x_1, x_2, s_1 \geq 0\end{array}$$

Another Example of Conversions

$$\begin{array}{ll}\text{maximize} & z = -5x_1 - 3x_2 + 7x_3 \\ \text{subject to} & 2x_1 + 4x_2 + 6x_3 = 7 \\ & 3x_1 - 5x_2 + 3x_3 \leq 5 \\ & -4x_1 - 9x_2 + 4x_3 \leq -4 \\ & x_1 \geq -2, \quad 0 \leq x_2 \leq 4, \quad x_3 \text{ free}\end{array}$$



$$x_1 = x'_1 - 2 \text{ and } x_3 = x'_3 - x''_3$$

$$\begin{array}{ll}\text{minimize} & z' = 5x'_1 + 3x_2 - 7x'_3 + 7x''_3 \\ \text{subject to} & 2x'_1 + 4x_2 + 6x'_3 - 6x''_3 = 11 \\ & 3x'_1 - 5x_2 + 3x'_3 - 3x''_3 + s_2 = 11 \\ & 4x'_1 + 9x_2 - 4x'_3 + 4x''_3 - e_3 = 12 \\ & x_2 + s_4 = 4 \\ & x'_1, x_2, x'_3, x''_3, s_2, e_3, s_4 \geq 0\end{array}$$

Another Example of Conversions

$$\begin{array}{ll}\text{maximize} & z = 3x_1 + 2x_2 - x_3 + x_4 \\ \text{subject to} & x_1 + 2x_2 + x_3 - x_4 \leq -5 \\ & -2x_1 - 4x_2 + x_3 + x_4 \leq -1 \\ & -4x_1 - 9x_2 + 4x_3 \leq -4 \\ & x_1 \geq 0, \quad x_2 \leq 0, \quad x_3, x_4 \text{ free}\end{array}$$



$$x_2 = -x_2' \quad \text{and} \quad x_3 = x_3' - x_3'' \quad \text{and} \quad x_4 = x_4' - x_4''$$

$$\begin{array}{ll}\text{minimize} & z' = -3x_1 + 2x_2' + x_3' - x_3'' - x_4' + x_4'' \\ \text{subject to} & -x_1 + 2x_2' - x_3' + x_3'' + x_4' - x_4'' - e_1 = 5 \\ & 2x_1 - 4x_2' - x_3' + x_3'' - x_4' + x_4'' - e_2 = 1 \\ & 4x_1 - 9x_2' - 4x_3' + 4x_3'' - e_3 = 4 \\ & x_1, x_2', x_3', x_3'', x_4', x_4'', e_1, e_2, e_3 \geq 0\end{array}$$

Part 3: Feasible Solutions and Basic Feasible Solutions



Feasible Solutions and Basic Feasible Solutions

- Recall from lecture 2 that feasible solutions satisfy the problem constraints

DEFINITION: A **basic solution** x is any point such that

1. It satisfies the equality constraints of the standard form ($Ax = b$)
2. The columns of A that correspond to nonzero values of x are linearly independent

DEFINITION: A **basic feasible solution** x is a basic solution that also satisfies the non-negativity constraint $x \geq 0$

The basic feasible solutions are the corners of the boundary of the feasible set

Theorem: A point x is an extreme point of the set $Ax = b$ with $x \geq 0$ if and only if it is a basic feasible solution.

Next we look at why this is helpful for solving problems.

Extreme Points in Geometry versus Basic Feasible Solutions

$$\begin{array}{ll} \text{minimize} & z = -x_1 - 2x_2 \\ \text{subject to} & -2x_1 + x_2 \leq 2 \\ & -x_1 + x_2 \leq 3 \\ & x_1 \leq 3 \\ & x_1 \geq 0, x_2 \geq 0 \end{array}$$

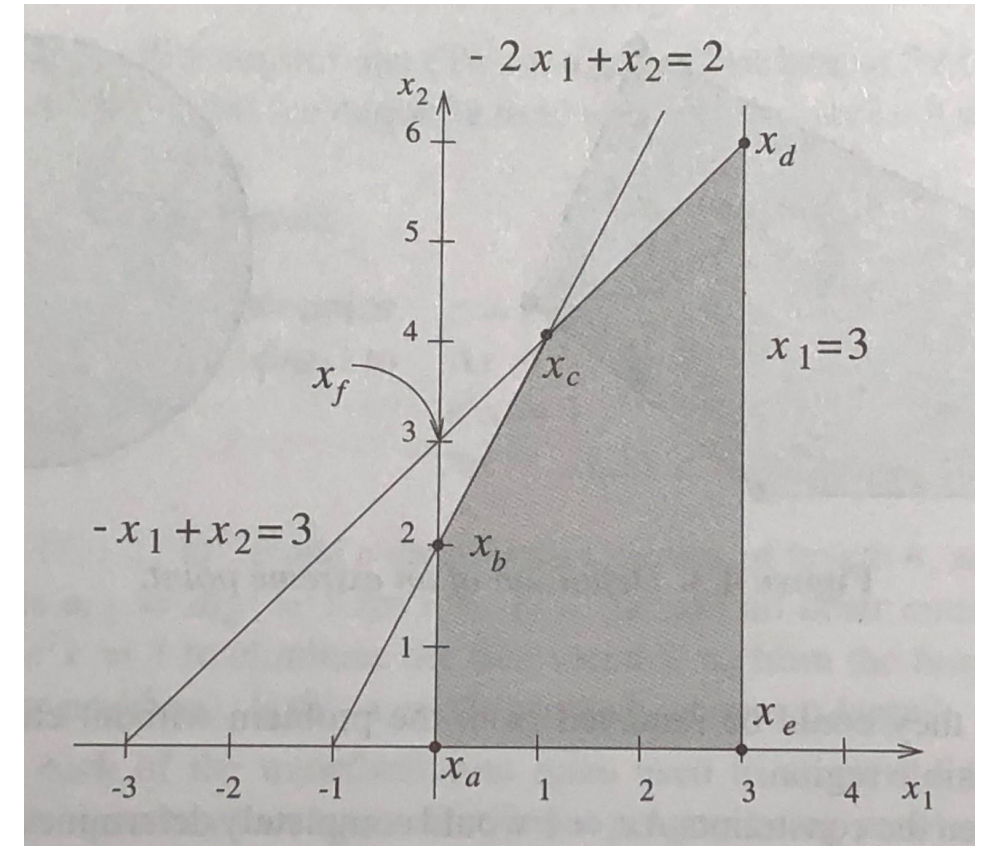
5 constraints lead to 10 potential intersecting corners

We only have 5, why?

A: 4 are outside, and 2 parallel ones never intersect

$$\begin{array}{ll} \text{minimize} & z = -x_1 - 2x_2 \\ \text{subject to} & -2x_1 + x_2 + s_1 = 2 \\ & -x_1 + x_2 + s_2 = 3 \\ & x_1 + s_3 = 3 \\ & x_1, x_2, s_1, s_2, s_3 \geq 0 \end{array}$$

$$Ax = b \Rightarrow \begin{bmatrix} -2 & 1 & 1 & 0 & 0 \\ -1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \\ s_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix}$$



White Board

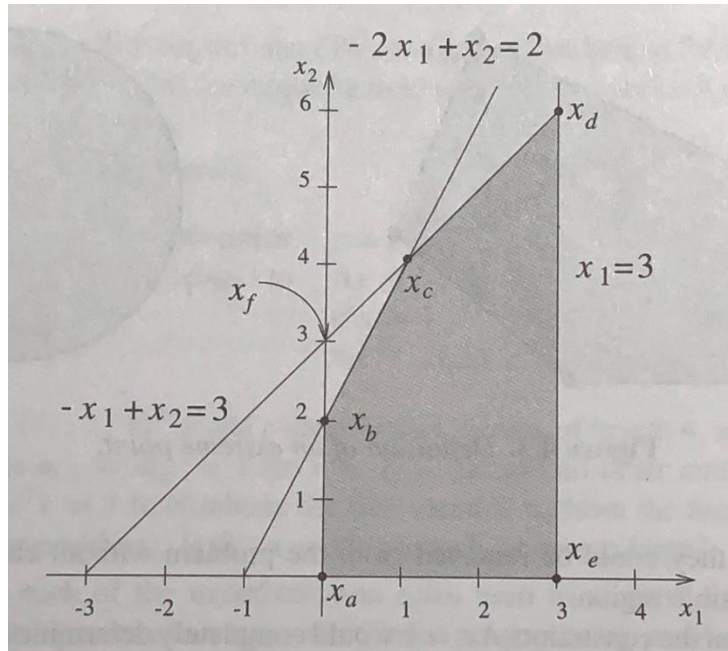
What are the Basic Feasible Solutions?

- Recall that the basic feasible solutions must meet $Ax = b$ and $x \geq 0$
 - Let's use some linear algebra to find some points that are basic solutions:
 - 1) Let A be $m \times n$ and pick m basic variables x_B (remainder are x_N which we set to zero)
 - 2) Rearrange $A = [B \ N]$ where B corresponds to the basic variables
 - 3) Now, $Ax = [B \ N] \begin{bmatrix} x_B \\ x_N \end{bmatrix} = Bx_B + Nx_N = Bx_B = b$
 - 4) Note this reduced the problem to $Bx_B = b$ and since B is invertible, we have $x_B = B^{-1}b$
 - 5) CAUTION: This will create some basic solutions that are NOT basic feasible solutions (i.e. some x may be negative)
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Finding the Basic Feasible Solutions

$$Ax = b$$

$$\begin{bmatrix} -2 & 1 & 1 & 0 & 0 \\ -1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \\ s_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix}$$



a) Try $x_B = [x_2 \ s_1 \ s_3]^T$ so that $B = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $B^{-1} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $x_B = B^{-1}b$

now, $x_B = [3 \ -1 \ 3]^T$ and thus $x = [0 \ 3 \ -1 \ 0 \ 3]^T$

This is point x_f on the plot → Not feasible

b) Try $x_B = [s_1 \ s_2 \ s_3]^T$ so that $B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $B^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $x_B = B^{-1}b$

now, $x_B = [2 \ 3 \ 3]^T$ and thus $x = [0 \ 0 \ 2 \ 3 \ 3]^T$

This is point x_a on the plot → Feasible

c) Try $x_B = [x_1 \ x_2 \ s_1]^T$ so that $B = \begin{bmatrix} -2 & 1 & 1 \\ -1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$, $B^{-1} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix}$, $x_B = B^{-1}b$

now, $x_B = [3 \ 6 \ 2]^T$ and thus $x = [3 \ 6 \ 2 \ 0 \ 0]^T$

This is point x_d on the plot → Feasible

Main Result

LINEAR PROGRAM THEOREM:

If a linear program in standard form has a finite optimal solution, then it has an optimal basic feasible solution.

Note that there may be an infinite optimal solution (if unbounded).

Also, if there are m basic variables and n total variables (ie: A is $m \times n$), then there are potentially a lot of basic feasible solutions, in fact the number can be as high as:

$$\binom{n}{m} = \frac{n!}{m! (n - m)!}$$

We thus could sequence through all of the basic feasible solutions to find the optimal result, but the Simplex Method will give us a more efficient way to go through them.

Summary

- I will post a PDF version of these slides on BrightSpace
 - Today we saw the geometry behind linear programming
 - An important part of this was putting problems into Standard Form – practice this on HW
 - Next week we will start to look at the Simplex Method to efficiently find the optimal point
 - HW #3 is due next week
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